

BIO503 (Biophysics)

Current File (Final Term - 2022)

We Grow Together

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Write down the function of cell membrane. 5 Marks

Functions of the cell membrane:

- To maintain the physical integrity of the cell that is to mechanically enclose the contents of the cell, and also to control the movement of particles e.g., ions of molecules, into and out of the cell.
- The cell membrane maintains the physical integrity of the cell.
- It's most obvious in the cases of animal cells (because they don't have cell walls) that the cell membrane holds the cell together by enclosing the cytoplasm and organelles within it.
- The cell membrane forms a barrier between the inside of the cell and the environment outside the cell enclosing cytoplasm and any organelles within the cell and enabling different chemical environments to exist on each side of the cell membrane.
- The cell membrane physically separates the intracellular components (e.g., organelles in eukaryotic cells) from the extracellular environment.
- The cell membrane protects the cell from some harmful chemicals in its external environment.
- It also protects the cell from loss of useful biological macromolecules held within the cell by its plasma membrane.

What is active and passive electrolocation? 5 Marks

Active electrolocation:

In active electrolocation, the animal senses its surrounding environment by generating electric fields and detecting distortions in these fields using electroreceptor organs. This electric field is generated by means of a specialized electric organ consisting of modified muscle or nerves. This field may be modulated so that its frequency and wave form are unique to the species and sometimes, the individual.

Passive electrolocation:

In passive electrolocation, the animal senses the weak bioelectric fields generated by other animals and uses it to locate them. These electric fields are generated by all animals due to the activity of their nerves and muscles. A second source of electric fields in fish is the ion pumps associated with osmoregulation at the gill membrane. This field is modulated by the opening and closing of the mouth and gill slits. Many fish that prey on electrogenic fish use the discharges of their prey to detect them. This has driven the prey to evolve more complex or higher frequency signals that are harder to detect.

What is electrolocation? 3 Marks

Electrolocation:

Electro-receptive animals use this sense to locate objects around them. This is important in ecological niches where the animal cannot depend on vision: for example, in caves, in murky water and at night. Many fish use electric fields to detect buried prey. Some shark embryos and pups "freeze" when they detect the characteristic electric signal of their predators.

Name of any four Amino acid?

- Histidine
- Isoleucine
- Leucine
- Lysine
- Methionine
- Phenylalanine
- Threonine
- Tryptophan
- Valine

What is Microsievert and millisievert?

Microsievert (MSV) is one millionth of a sievert ($1/1,000,000$)

Millisievert (mSv) is one thousandth of a sievert ($1/1,000$)

What is Saxitoxin?

Saxitoxin is a potent neurotoxin and the best-known paralytic shellfish toxin. Ingestion of saxitoxin by humans, usually by consumption of shellfish contaminated by toxic algal blooms, is responsible for the illness known as paralytic shellfish poisoning.

Name phases of matter?

- Solid
- Liquid
- Gas

Name the groups of Ion channel? 3 Marks

- Ligand gated channels (neurotransmitters)
- Voltage gated channels trans-membrane potential (electric field)
- Second messenger gated channels nucleotides, (G-proteins)
- Mechano-sensitive channels osmotic pressure (membrane curvature)
- Gap junctions, (porins not gated)

What is Spectroscopy?

Spectroscopy is the study of the interaction between matter and electromagnetic radiation. Spectroscopy relates to the dispersion of an object's light into its component colours (i.e. Energies). Spectroscopy and spectrography are terms used to refer to the measurement of radiation intensity as a function of wavelength and are often used to describe experimental spectroscopic methods.

State two types of self-assembly? 2 Marks

There are two types of self-assembly. These are

1. Intra-molecular self-assembly
2. Intermolecular self-assembly

State fluid mosaic model? 2 Marks

The fluid mosaic model describes the structure of the plasma membrane as a mosaic of components including phospholipids, cholesterol, proteins and carbohydrates that gives the membrane a fluid character.

What is Radiation Protection? 2 Marks

Radiation protection, sometimes known as radiological protection, is defined by the International Atomic Energy Agency (IAEA) as "The protection of people from harmful effects of exposure to ionizing radiation, and the means for achieving this".

Who discovered DNA? And name of scientist discovered the structure of DNA?

2 Marks

In 1953 Watson and Crick determined the structure of DNA.

Define HPLC?

High-performance liquid chromatography (HPLC) formerly referred to as high-pressure liquid chromatography is a technique in analytical chemistry used to separate, identify, and quantify each component in a mixture. High performance liquid chromatography is powerful tool in analysis, it yields high performance and high speed compared to traditional columns chromatography. HPLC is a chromatographic technique that can separate a mixture of compounds.

Define MRI (Magnetic Resonance Imaging)?

A medical imaging technique which is used in radiology to form images of the anatomy and the physiological processes of the body in both health and disease.

Define Electrophoresis?

Electrophoresis separates mixtures of proteins based on charge, charge/mass ratio or size.

State Na⁺K⁺ATP function?

Define Translatory motion and its types?

In Translatory motion, the particle moves from one point in space to another. This motion may be along a straight line or along a curved path.

Types of Translatory motion :

Rectilinear Motion: Motion along a straight line is called rectilinear motion.

Curvilinear Motion: Motion along a curved path is called curvilinear motion.

State Animal use aquatic sound?

Some usually aquatic animals such as sharks have acute bioelectric sensors providing a sense known as electroreception while migratory birds navigate in part by orienteering with respect to the Earth's magnetic field.

What is Biophysics ?

Biophysics is an interdisciplinary science that applies the approaches and methods of physics to study biological systems. Biophysical research shares significant overlap with biochemistry, physical chemistry, nanotechnology, bioengineering, computational biology, biomechanics and systems biology.

Three applications of Ion Exchange Chromatography?

- Used in the analysis of amino acids. The amino acid "Auto analyser" is based on ion exchange principle.
- Determine the base composition of nucleic acids. Chargaff used this technique for established the equivalence of Adenine and Thymine; Guanine and Cytosine.
- This is most effective method for water purification. Complete deionization of water (or) a non-electrolyte solution is performed by exchanging solute cations for hydrogen ions and solute anions for hydroxyl ions. This is usually achieved by method is used for softening of drinking water. Proteins are also successfully separated by this technique.

What is Metabolic pathway?

A metabolic pathway is a series of chemical reactions in a cell that build and breakdown molecules for cellular processes. Most metabolic pathways take place in specific regions of the cell. Anabolic pathways synthesize molecules and require energy. Catabolic pathways break down molecules and produce energy. Two types of metabolic pathway occur in a cell

- Catabolic pathway
- Anabolic pathway

Protein isolation and purification?

Protein isolation and purification achieved in several steps namely,

- Cell disruption

- Fractionation
- Crude sample preparation
- Purification

What is immune-blotting?

Immunoblotting is a protein visualization technique applied in combination with affinity chromatography.

Define Photosynthesis?

Most important transfer event, for life on our planet is the biological process called photosynthesis, which occurs in most plants, algae, and some bacteria. Photosynthesis is the process wherein light energy from the sun is converted into chemical energy to make organic compounds, or food (mainly sugars), for the organism.

What is Plant bioacoustics?

Measured sound emissions by plants as well as differential germination rates, growth rates and behavioural modifications in response to sound are well documented. Plants detect neighbours by means other than well-established communicative signals including volatile chemicals, light detection, direct contact and root signalling because sound waves travel efficiently through soil and can be produced with minimal energy expenditure, plants may use sound as a means for interpreting their environment and surroundings.

Define energy?

Energy is very important in our life, where we can't carry out different activities (mental or muscular) without the produced energy from burning sugar inside our bodies, we can't cook our food without the heat energy produced from the burning of natural gas. There are various forms of energy like Chemical energy, Heat energy, Light energy, Electrical energy and Kinetic energy.

What is SPECT?

SPECT is a nuclear imaging scan that integrates computed tomography (CT) and a radioactive tracer. The tracer is what allows doctors to see how blood flows to tissues and organs. Before the SPECT scan, a tracer is injected into your bloodstream.

Define Radioactive decay?

It is a process by which the nucleus of an unstable atom loses energy by emitting radiation is known as radioactive decay in which radiation is released in the form of alpha particles, beta particles, gamma rays.

What is Biophysical techniques?

Biophysical techniques provide information about the electronic structure, size, shape, dynamics, polarity, and modes of interaction of biological molecules. Some of the most

exciting techniques provide images of cells, subcellular structures, and even individual molecules.

Direct effect ionizing radiation?

When cells are exposed to ionizing radiation, radiochemical damage can occur either by direct action or indirect action. Direct action occurs when alpha particles, beta particles or x-rays create ions which physically break one or both of the sugar phosphate backbones or break the base pairs of the DNA.

What is reverse Phase HPLC?

The stationary phase is nonpolar (hydrophobic) in nature, while the mobile phase is a polar liquid, such as mixtures of water and methanol or acetonitrile. It works on the principle of hydrophobic interactions hence the more nonpolar the material is, the longer it will be retained.

What are the Components Of Gel Filtration?

The basis components of Gel filtration experiment are:

- Matrix
- Chromatography column
- Elution buffer

Define Electro-dialysis?

Electro dialysis (ED) is used to transport salt ions from one solution through ion-exchange membranes to another solution under the influence of an applied electric potential difference.

Write the Instruments of NMR?

Sample holder : Glass tube with 8.5 cm long, 0.3 cm width.

Permanent magnet: It provides homogeneous magnetic field at 60-100 MHz.

Magnetic coils : These coils induce magnetic field when current flows through them.

Sweep generator: To produce the equal amount of magnetic field pass through the sample.

Radio frequency (radio transmitter coil) : A radio transmitter coil that produces a short powerful pulse of radio waves.

Radio frequency (radio receiver coil): A radio receiver coil that detects Receiver radio frequencies.

Formation Of enthalpy?

The enthalpy of formation of a substance is defined as the heat change i.e. heat evolved or absorbed when 1 mole of the substance is formed from its elements under given conditions of temperature and pressure. It is usually represented by $\Delta_f H$.

What is Chromatography?

The term “chromatography” is derived from Greek, chroma meaning, “colour,” and graphein meaning “to write.” Chromatography is the technique for the separation, purification, and testing of compounds.

What is RNA and its types?

RNA :

RNA molecules are single-stranded nucleic acids composed of nucleotides. RNA plays a major role in protein synthesis as it is involved in the transcription, decoding, and translation of the genetic code to produce proteins. RNA stands for ribonucleic acid and like DNA, RNA nucleotides contain three components:

- A Nitrogenous Base
- A Five-Carbon Sugar
- A Phosphate Group

Types of RNA :

- messenger RNA (mRNA)
- transfer RNA (tRNA)
- ribosomal RNA (rRNA)

What is Mass spectrometry?

Mass spectrometry (MS) is an analytical technique that is used to measure the mass-to-charge ratio of ions.

Write short note on Structure of RNA?

RNA consists of four nitrogenous bases: adenine, cytosine, uracil, and guanine. Uracil is a pyrimidine that is structurally similar to the thymine, another pyrimidine that is found in DNA. Like thymine, uracil can base-pair with adenine.

DNA location in prokaryotes?

The DNA in prokaryotes is contained in a central area of the cell called the nucleoid, which is not surrounded by a nuclear membrane. Many prokaryotes also carry small, circular DNA molecules called plasmids, which are distinct from the chromosomal DNA and can provide genetic advantages in specific environments.

History of biophysics? 5 Marks

Biophysics is a relatively young discipline but its origins go back a long way. The term biophysics was originally introduced by Karl Pearson in 1892. The term biophysics is also regularly used in academia to indicate the study of the physical quantities (e.g. electric current, temperature, stress, entropy) in biological systems.

1943: The popularity of biophysics rose sharply when published a book "What is Life": Erwin

Schrodinger 1946: Biophysics research Unit, King's College, London, hire physicists to work on questions of biological significance; Maurice Wilkins, Rosalind Franklin: X-ray diffraction of DNA

1953 Francis Crick (particle physicist turned into biophysicist at Cambridge) and James Watson (biologist): double helix structure of DNA

1957: The American Biophysics Society was founded

1959: A series of conferences on Biophysics was organized in Genoa and in other cities. It was considered extremely novel for physics scientists to have common interests with medicine and biology scientists. The first National Biophysics Congress was held from 3-5 June, 1963 in Italy.

Define Thermometer?

A thermometer is a device that measures temperature or a temperature gradient (the degree of hotness or coldness of an object).

Define DNA microarrays?

A DNA microarray is a collection of microscopic DNA spots attached to a solid surface. Scientists use DNA microarrays to measure the expression levels of large numbers of genes simultaneously or to genotype multiple regions of a genome. Each DNA spot contains Pico moles of a specific DNA sequence, known as probes.

Define Equilibrium?

When the rates of the forward and reverse reactions become equal, the concentrations of the reactants and the products remain constant.

What is Catabolic process?

Catabolism is the branch of the metabolic process that breaks down complex, large molecules into smaller ones, yielding energy. It is the destructive branch of the metabolism that results in the release of energy. Each living cell depends on energy for its existence.

What are the types of Ultra-centrifugation?

There are two types of Ultra-centrifugation

- Analytical ultracentrifuge (AUC)
- Preparative ultracentrifuge

What is Ultra-centrifugation?

Ultracentrifuge is a sophisticated and advanced centrifuge that operates at an extremely high speed and separates smaller molecules that cannot be separated from the traditional centrifuges.

Difference between kinetic energy and potential energy?

Kinetic Energy

- Kinetic energy is the kind of energy present in a body due to the property of its motion.
- It can be easily transferred from one body to another.
- The determining factors for kinetic energy are Speed or velocity and mass.
- Flowing water is one of the examples of kinetic energy.
- It is relative with respect to nature.

Potential Energy

- Potential Energy is the type of energy present in a body due to the property of its state.
- It is not transferable.
- Here, the determining factors are Height/ distance and mass.
- Water present at the top of a hill is an example of potential energy.
- It is non-relative with respect to nature.

Explain 3rd law of thermodynamics? Taking water as examples? 5 marks

The third law of thermodynamics states that the entropy of a perfect crystal at a temperature of zero Kelvin (absolute zero) is equal to zero. The Third Law of Thermodynamics can be envisioned by taking water as an example. Water in gas form has molecules that can move around very freely. Water vapour has very high entropy (randomness). As the gas cools, it becomes liquid. The liquid water molecules can still move around, but not as freely. They have lost some entropy. When the water cools further, it becomes solid ice. The solid water molecules can no longer move freely, but can only vibrate within the ice crystals. The entropy is now very low. As the water is cooled more, closer and closer to absolute zero, the vibration of the molecules diminishes. If the solid water reached absolute zero, all molecular motion would stop completely. At this point, the water would have no entropy (randomness) at all.

What is Enthalpy and Entropy?

Enthalpy is the measure of total heat present in the thermodynamic system where the pressure is constant. It is represented as $\Delta H = \Delta E + P \Delta V$ where E is the internal energy.

Entropy is the measure of degree of disorder or randomness in a thermodynamic system.

Difference between intermolecular and intramolecular forces

Intramolecular forces are the forces that hold atoms together within a molecule.

Examples of intramolecular forces

- Ionic bonds

- Hydrogen Bonding
- Van der Waals dipole-dipole interactions
- Van der Waals dispersion forces.

Intermolecular forces are forces that exist between molecules.

Examples of intermolecular forces

- London dispersion force
- Dipole-dipole interaction
- Ion-dipole interaction
- Van der Waals forces

What is Ion- dipole force?

An ion- dipole force is an attractive force that results from the electrostatic attraction between an ion and a neutral molecule that has a dipole. Most commonly found in solutions. Especially important for solutions of ionic compounds in polar liquids.

- A positive ion (cation) attracts the partially negative end of a neutral polar molecule.
- A negative ion (anion) attracts the partially positive end of a neutral polar molecule

Define covalent bond?

A covalent bond also called a molecular bond, is a chemical bond that involves the sharing of electron pairs between atoms.

Define Salting Out?

Salting out is a purification technique that utilizes the reduced solubility of certain molecules in a solution of very high ionic strength. Salting out is typically used to precipitate large biomolecules, such as proteins or DNA.

What is Friction and cause of Friction?

Friction :

Friction is a force that resists the relative motion between two objects or materials.

Cause of Friction :

- Interlocking of irregularities.
- Chemical bonding.
- Electrostatic interaction.
- Electromagnetic interaction.
- Pressure

Define solution?

The mixture of a solute in a solvent is called a solution.

What is Hypertonic , Hypotonic and Isotonic?

Hypertonic :

The solution with the higher solute concentration is hypertonic.

Hypotonic :

The solution with the lower solute concentration is hypotonic.

Isotonic :

Solutions of equal solute concentration are isotonic.

Difference between velocity and speed?

Velocity :

Velocity is the rate and direction of an object's movement. Velocity is a vector quantity.

Speed :

Speed is the time rate at which an object is moving along a path. Speed is a scalar quantity.

Write four polar & non-polar amino acids?

Polar Amino Acids :

Polar amino acids are hydrophilic.

Examples :

Arginine, Asparagine, Aspartic acid (or aspartate), Glutamine, Glutamic acid (or glutamate), Histidine, Lysine, Serine and Threonine.

Non-Polar Amino Acids :

Non-polar amino acids are hydrophobic.

Examples :

Alanine, Cysteine, Glycine, Isoleucine, Leucine, Methionine, Phenylalanine, Proline, Tryptophan, Tyrosine and Valine.

Define isotopes?

Isotopes are the atoms of an element which have the same number of protons but different numbers of neutrons.

For example, carbon-14, carbon-13, and carbon-12 are all isotopes of carbon.

Name types of intermolecular forces that exist between molecules?

- London dispersion forces (LDF)
- Dipole- dipole interactions
- Hydrogen bonding
- Ion dipole forces

What is Ultra-filtration?

The process of separating colloidal particles from the soluble solutes(impurities) using specially prepared filters, which are permeable to all substances except the colloid.

What is NMR?

Nuclear magnetic resonance spectroscopy, most commonly known as NMR spectroscopy, is a research technique that exploits the magnetic properties of certain atomic nuclei. This type of spectroscopy determines the physical and chemical properties of atoms or the molecules in which they are contained. It relies on the phenomenon of nuclear magnetic resonance and can provide detailed information about the structure, dynamics, reaction state, and chemical environment of molecules.

Write chemical structure of DNA?

Each strand of a DNA molecule is composed of a long chain of monomer nucleotides. The nucleotides of DNA consist of a deoxyribose sugar molecule to which is attached a phosphate group and one of four nitrogenous bases: two purines (adenine and guanine) and two pyrimidines (cytosine and thymine).

What is chiral phase of liquid crystal?

Cholesteric liquid crystals have the molecules in a twisted and chiral arrangement. The term cholesteric comes from its first observation where this phase of matter first appeared in cholesterol derivatives. Have 360 degree twisting and chirality.

Difference between Nematic and Sematic phase/crystals?

Nematic Crystal Liquid:

- Nematic crystals have no ordered structure of molecules.
- The term Nematic comes from a Greek origin where it has the meaning "Thread".
- Molecules tend to align in a single direction Upon the application of external magnetic or electric field.

Sematic liquid crystals

- Sematic liquid crystals have a layered molecular structure.
- The term Sematic originates from the Latin word 'semectius' which means "cleaning" ;in other words, it means the presence of soap-like properties. Therefore, the Sematic phase is positioned in order along one direction.
- Have 360 degree twisting and chirality.

What is diffusion?

Diffusion is the movement of molecules from an area of high concentration to low concentration.

Write down the 2 functions of proteins?

1. Protein is termed the building block of the body. It is called this because protein is vital in the maintenance of body tissue, including development and repair. Hair, skin, eyes, muscles and organs are all made from protein.
2. Protein is a major source of energy.
3. Protein is involved in the creation of some hormones. These substances help control body functions that involve the interaction of several organs.
4. Enzymes are proteins that increase the rate of chemical reactions in the body.
5. Protein is a major element in transportation of certain molecules. For example, haemoglobin is a protein that transports oxygen throughout the body.
6. Protein forms antibodies that help prevent infection, illness and disease.

Write about Kelvin and Celsius scale?

Kelvin Scale :

A temperature scale having an absolute zero below which temperatures do not exist.

Celsius scale

Celsius scale is also called the centigrade scale because the scale is divided into 100 divisions. The freezing point of water on this scale is 0 degrees Celsius i.e. 0°. The boiling point of water is 100 degrees Celsius i.e. 100°.

Explain factors that make membrane fluid in nature?

Factors :

The mosaic nature of the membrane, its phospholipid chemistry, and the presence of cholesterol contribute to membrane fluidity.

- THE LENGTH OF THE FATTY ACID TAIL
- TEMPERATURE
- CHOLESTEROL CONTENT OF THE BILAYER
- THE DEGREE OF SATURATION OF FATTY ACIDS TAILS

What is dialysis?

Dialysis is the process of separating molecules in solution by the difference in their rates of diffusion through a semipermeable membrane.

Research biophysics in future?

What is dissociation constant?

Dissociation constant is a type of equilibrium constant that specifically involves the measure of the propensity of dissociation of a complex molecule into its

subcomponents. The dissociation constant is sometimes called ionization constant when applied to salts.

Example :

How tightly a ligand binds to a particular protein.

Define Self-assembly?

Self-assembly is the process of association of individual units of a material into highly arranged/ordered structures/patterns.

Write down the plant and animal bioacoustics?

Explain molecule motor?

Molecular motors are a class of proteins that drive intracellular trafficking by converting chemical energy to mechanical work along cytoskeletal filaments. There are two major families of molecular motors that transport organelles throughout the cell. These families include the dynein family and the kinesin family. Both have very different structures from one another and different ways of achieving a similar goal of moving organelles around the cell.

Difference between homonuclear or heteronuclear diatomic molecules?

Homonuclear diatomic molecules

- Homonuclear diatomic molecules are substances composed of two atoms of the same chemical element bonded to each other via covalent bonds.
- Stable isotopes are very stable and do not undergo radioactive decay.
- Homonuclear diatomic molecules have nonpolar covalent bonds.
- Homonuclear diatomic molecules have identical atoms.

Heteronuclear diatomic molecules

- Heteronuclear diatomic molecules are substances composed of two atoms of two different chemical elements bonded to each other via covalent bonds.
- Heteronuclear diatomic molecules have atoms of different chemical elements.
- Heteronuclear diatomic molecules have polar covalent bonds.
- Heteronuclear diatomic molecules have different atoms.

Write short note on Electro-dialysis?

Electro dialysis (ED) is used to transport salt ions from one solution through ion-exchange membranes to another solution under the influence of an applied electric potential difference.

What is isotropic?

Isotropic materials are materials whose properties remain the same when tested in different directions. Isotropic materials differ from anisotropic materials, which display

varying properties when tested in different directions. Common isotropic materials include glass, plastics, and metals.

Or

Properties of a material are identical in all directions.

What is anisotropic?

Properties of a material depend on the direction.

For Example:

Wood, composite materials, all crystals etc.

What are the factors that Affect Lattice Energy?

The two primary factors that affect the lattice energy of an ionic compound

- The charges on the ions
- The radius of the ions

Explain lattice enthalpy?

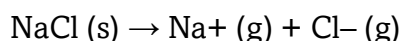
Lattice enthalpy is a measure of the strength of the forces between the ions in an ionic solid. The greater the lattice enthalpy, the stronger the forces. Those forces are only completely broken when the ions are present as gaseous ions, scattered so far apart that there is negligible attraction between them.

Factors that Affect Lattice Energy :

- The charges on the ions
- The radius of the ions

Example

The crystal lattice of a sodium chloride molecule. In the case of this ionic molecule, the lattice energy is the energy required for the following reaction to proceed.



Here, the energy that must be supplied to 1 mole of sodium chloride to separate it into gaseous Na^+ and Cl^- ions is 786 kilojoules.

Define physics ?

A physics is a science that deals with matter and energy and their interactions.

What is osmosis?

Osmosis is the movement of water from an area of lower concentration to an area of higher concentration.

State the cofactor, coenzyme, prosthetic group and active site?

Cofactors

Cofactors that increase the rate of reaction or are required for enzyme function. Cofactors are not proteins but rather help proteins, such as enzymes, although they can also help non-enzyme proteins as well. Examples of cofactors include metal ions like iron and zinc.

Coenzymes

A specific type of cofactor, coenzymes, are organic molecules that bind to enzymes and help them function.

Prosthetic Groups

Prosthetic groups are cofactors that bind tightly to proteins or enzymes.

Active Sites

The active site is the region on the enzyme where the substrate binds.

Why cell membrane is called as a selectively membrane?

The cell membrane is a very thin layer of protein and fat. It allows only selective substances to pass through it, hence, it is called a selectively permeable membrane.

Describe the structure of DNA?

The DNA molecule consists of two strands that wind around one another to form a shape known as a double helix. Each strand has a backbone made of alternating sugar (deoxyribose) and phosphate groups. Attached to each sugar is one of four bases-- adenine (A), cytosine (C), guanine (G), and thymine (T).

What is sedimentation?

It is defined as the separation process in which solids are separated from the liquid. All the solids settle down at the bottom of a beaker and on top, a clear layer of liquid is obtained. Sedimentation is a process by which heavier impurities present in liquid normally water settle down at the bottom of the container containing the mixture. The process takes some amount of time.

Example

Water treatment plants use the method of sedimentation to filter out unwanted particles from unclean water.

State the Bourdon tube, pressure transducer and barometer?

Bourdon Tube is a device that measures pressure using mechanical deformation.

Pressure Transducers are devices that use piezoelectrics to measure pressure very accurate and robust and can measure Pgauge or Pabs.

Barometer is a device that measures atmospheric pressure. It is a manometer with a near vacuum on one end.

What is osmotic pressure? 2 Marks

Osmotic pressure is the minimum pressure which needs to be applied to a solution to prevent the inward flow of its pure solvent across a semipermeable.

What is dissociation constant in water?

When ionic compounds dissolve in water, their ions separate from one another in a process called dissociation. The value of the dissociation constant of water, K_w , is 1.0×10^{-14} .

Explain chiral phase in liquid crystal?

Cholesteric liquid crystals have the molecules in a twisted and chiral arrangement. The term cholesteric comes from its first observation where this phase of matter first appeared in cholesterol derivatives. Have 360 degree twisting and chirality. The thickness of the chiral phase is such that polarized light passing through it is rotated by 90° , which corresponds to the orientation of the right-hand polarizing filter. So in this state, the light passes through and illuminates the pixel surface.

Name four common solvent other than water . 2 marks.

- Ethanol
- Methanol
- Ethyl Acetate
- Hexanes

What is animal sound acoustics? 2 marks.

What is macromolecular assembly? 5 marks

The term macromolecular assembly (MA) refers to massive chemical structures such as viruses and non-biologic nanoparticles, cellular organelles and membranes and ribosomes, etc. that are complex mixtures of polypeptide, polynucleotide, polysaccharide or other polymeric macromolecules.

What is DNA microarray? 3 marks

A microarray is a laboratory tool used to detect the expression of thousands of genes at the same time. DNA microarrays are microscope slides that are printed with thousands of tiny spots in defined positions, with each spot containing a known DNA sequence or gene.

Applications of DNA microarray? 5 Marks

- To study transcriptomes and proteome
- To diagnose pathogenic as well as genetic diseases in man
- To identify microbes in the environment with the help of species-specific probes
- To genotype genomes through single nucleotide polymorphism (SNP) analysis
- To detect gene expression of mRNAs of a particular cell at different times
- To measure changes in the level of gene expression
- To observe DNA mutations

- To study genomic gains and losses

What is the future of biophysics? 5 marks

As such this is almost impossible to answer. I did not major in biophysics. My bachelor's degree is in biotechnology. No one can predict future. So long Biology and Physics exist, Biophysics will be there as a subject with its challenges. Biophysics is a field of applied physics that concerns itself in studying the most complex soft matter there is, that is biological matter and understanding the physical properties that distinguish it from otherwise non-living matter and studies how physical phenomena and concepts can be applied to biological and/or medical, to a greater extend.

Ek numerical aya tha 5 marks ka jis ma temperature find krna tha Celsius by spectrometer....* $\Delta H = 155\text{KJ/mol}$ and $\Delta S = 345\text{KJ/mol}$.*

Given :

$$\Delta H = 155\text{KJ/mol}$$

$$\Delta S = 345\text{KJ/mol}$$

T = Find

$$\Delta G = ?$$

Formula :

$$\Delta G = \Delta H - T\Delta S$$

Solve :

$$\Delta G = 155\text{KJ} - T(345\text{KJ})$$

We know that $\Delta G = \Delta H - T\Delta S$. Solving for ΔS , we have:

Not completed

Differentiate between Endothermic and Exothermic? 5 Mark

Exothermic reaction

- Reactants have more free energy than the products.
- Involve a net release of energy and or an increase in entropy.
- Occur spontaneously (without a net input of energy).

Endothermic reaction

- Reactants free energy than the products.
- Involve a net input of energy and/or a decrease in entropy.
- Do not occur spontaneously

What is Decantation and filtration?

Decantation

It is defined as the separation process in which two immiscible liquids are separated. This is done by pouring out the clear upper layer of liquid. Separation of a mixture of oil from water is an example of decantation.

Filtration

It is defined as the separation process in which insoluble impurities are separated from the solution. Filtration is done with the help of filter paper. Filtering mud from muddy water is an example of filtration.

Methods to reduce friction?

- Friction can be reduced by making the surface smooth by polishing
- Friction can be reduced by applying lubricants (like oil or grease) to the rubbing surface
- Friction can be reduced by using wheels to move objects
- Friction can be reduced by using ball bearing between the moving parts of machine.

Write the diffusion in plants and animals?

Diffusion in Plants

Plants have no special organs for breathing. They have to rely on diffusion for their supplies of oxygen and carbon dioxide. There are pores in the leaves and stems through which the gases diffuse. In daylight, CO₂ (for photosynthesis) will be diffusing in and O₂ will be diffusing out. In darkness, O₂ will diffuse in and CO₂ will diffuse out as a result of respiration. The plant is respiring all the time, using up oxygen and producing carbon dioxide. In daylight, photosynthesis will use up the carbon dioxide as fast as it is produced by respiration, SO there will be no outward diffusion of carbon dioxide. Similarly, the oxygen produced by photosynthesis is used up by respiration.

Diffusion in Animals

Diffusion is important to organisms because it is the process by which useful molecules enter the body cells and waste products are removed. Digested food molecules (amino acids, glucose) move down a concentration gradient from the intestine to the blood. Waste products such as carbon dioxide or urea travel by diffusion from body cells into the bloodstream. Oxygen moves from high concentration (in the air sac) to a lower concentration (in the blood). Carbon dioxide moves from high concentration (in the blood) to a lower concentration (in the air sac).

What is Voltage-gated ion channels?

Voltage-gated ion channels are a class of trans membrane proteins that form ion channels that are activated by changes in the electrical membrane potential near the channel.

Detail note on Endocytosis?

Endocytosis is the process in which cells absorb molecules by engulfing them. The plasma membrane creates a small deformation inward called an invagination, in which the substance to be transported is captured. There are two main kinds of endocytosis: Phagocytosis, or cellular eating, occurs when the dissolved materials enter the cell. The plasma membrane engulfs the solid material, forming a phagocytic vesicle. Pinocytosis, or cellular drinking, occurs when the liquid droplets are ingested by living cells.

Two major techniques of fractionation for fractionation of natural samples. 2 marks

Two main methods for fractionation of natural samples are as:

- Bioassay-guided fractionation
- Blood fractionation

Write down the direct effects of ionizing radiation?

There are two different types of damaging biological effects due to ionizing radiation on the cellular level.

- Exposure can cause either cell death.
- It can change its genetic information through mutations, leaving the cell able to reproduce itself.

Which one have higher boiling point H₂O or N₂?

The boiling point of H₂O is hundreds of degrees greater than the boiling point of N₂ because of stronger molecular interactions in H₂O (liq) than in N₂ (liq). The forces between molecules in H₂O(liq) are greater than those in N₂ (liq).

What are Amino acids?

Amino acids are the building blocks of proteins and are extremely important in biology. 20% of the human body is made up of protein. Protein plays a crucial role in almost all biological processes and amino acids are the building blocks of it. The chain of amino acids then folds into a unique shape that shape allows the enzyme to carry out specific chemical reactions.

Define Kinetic friction?

Sliding friction occurs between objects as they slide against each other. It acts in the direction opposite to the direction of motion. It prevents the object from moving too fast. It is also called as kinetic friction.

Define Dry Friction. 2 Marks

Dry friction resists relative lateral motion of two solid surfaces in contact.

- The two regimes of dry friction are 'static friction' ('stiction' between non-moving surfaces).
- Friction (sometimes called sliding friction or dynamic friction) between moving surfaces.

Explain ELISA? 3 Marks

In ELISA (enzyme- linked immunosorbent assay), the antigen is bound to a surface and then recognized by a specific antibody. Subsequently, a second antibody is applied that is specific to the first one and usually carries a covalently- bound enzyme molecule. By the addition of the substrate of the enzyme, a colourful and/or fluorescence. This way, the presence of the antigen or even its concentration can be determined. A more sophisticated variant of ELISA is the "sandwich" ELISA.

What are Proteins? 2 Marks

Proteins highly complex substance that is present in all living organisms. Proteins are of great nutritional value and are directly involved in the chemical processes essential for life.

Predict which will have the higher boiling point: N₂ or CO. Explain your reasoning. 5 Marks

CO and N₂ are both diatomic molecules with masses of about 28 amu, so they experience similar London dispersion forces. Because CO is a polar molecule, it experiences dipole-dipole attractions. Because N₂ is nonpolar, its molecules cannot exhibit dipole-dipole attractions. The dipole-dipole attractions between CO molecules are comparably stronger than the dispersion forces between nonpolar N₂ molecules, so CO is expected to have the higher boiling point.

Difference between Active and passive transport?

Active transport

Active transport moves molecules and ions from lower concentration to higher concentration with the help of energy in the form of ATP.

Passive transport

Passive transport moves molecules and ions from a higher concentration to lower concentration without any energy.

Write down the types of Frictions?

There are four types of friction: static, sliding, rolling, and fluid friction. Static, sliding, and rolling friction occur between solid surfaces. Fluid friction occurs in liquids and gases. All four types of friction are described below.

Static Friction

Static friction acts on objects when they are resting on a surface.

Sliding Friction

Sliding friction is friction that acts on objects when they are sliding over a surface. Sliding friction is weaker than static friction.

Rolling Friction

Rolling friction is friction that acts on objects when they are rolling over a surface. Rolling friction is much weaker than sliding friction or static friction.

Fluid Friction

Fluid friction is friction that acts on objects that are moving through a fluid.

Describe the hydrogen bonding in following?

Hydrogen Bonding in Proteins

Hydrogen bonds are a major feature of protein structure. They occur whenever a proton is shared by two electronegative atoms. Hence, only hydrogens bonded to nitrogen and oxygen atoms are usually considered in analyses of protein hydrogen bond networks.

Hydrogen Bonding in Plants

Hydrogen bonds are important to water movement in plants because they allow water to climb up through the roots and vascular system of the plant.

Hydrogen Bonds in DNA

DNA has a double-helix structure because hydrogen bonds hold together the base pairs in the middle. Hydrogen bonds are not chemical bonds. They can be easily disrupted. Without hydrogen bonds, DNA would have to exist as a different structure. This permits the DNA strands to separate for transcription and replication.

Why does a hydrogen bond occur?

The reason hydrogen bonding occurs is because the electron is not shared evenly between a hydrogen atom and a negatively charged atom. Hydrogen in a bond still only has one electron, while it takes two electrons for a stable electron pair. Any compound with polar covalent bonds has the potential to form hydrogen bonds.

Hydrogen bonding in water molecule?

The hydrogen bond in water is a dynamic attraction between neighbouring water molecules involving one hydrogen atom located between the two oxygen atoms. Hydrogen bonding forms in liquid water as the hydrogen atoms of one water molecule are attracted towards the oxygen atom of a neighbouring water molecule.

What is Buzz Pollination?

Buzz pollination is a type of pollination in which bees use vibrations to remove and collect pollen from flowers incidentally fertilising them.

Give examples of diffusion from our daily life?

- The smell of perfumes/Incense Sticks
- Opening the Soda/Cold Drinks bottle and the CO_2 diffuses in the air
- Dipping the tea bags in hot water will diffuse the tea in hot water
- Small dust particles or smoke diffuse into the air and cause air pollution
- Breathing and taking oxygen into the body and diffuses in our blood

What is Borromean rings?

Borromean rings are three simple closed curves in three-dimensional space that are topologically linked and cannot be separated from each other, but that break apart into two unknotted and unlinked loops when any one of the three is cut or removed.

What are the types of Thermodynamics properties?

Thermodynamic properties are divided into two broad types:

Intensive properties

Intensive properties are those that are independent of the size of a system such as Temperature, Pressure, Density. They are Independent of mass.

Here is a list of intensive properties.

- Colour
- Odour
- Lustre
- Ductility
- Conductivity
- Hardness and softness
- Density
- Pressure
- Malleability

Extensive properties

Extensive properties are those whose values depend on the size-or extent-of the system such as Mass m , Volume V , Total energy E . These are examples of extensive properties. If mass is increased, the value of extensive property also increases

Below is a list of extensive properties:

- Mass
- Weight
- Volume
- Length

Is heat Q the same thing as temperature T ?

Absolutely not. This is one of the most common misconceptions when dealing with the first law of thermodynamics. The heat Q represents the heat energy that enters a gas e.g. thermal conduction through the walls of the container. The temperature T on the other hand, is a number that's proportional to the total internal energy of the gas. So, Q is the energy a gas gains through thermal conduction, but T is proportional to the total amount of energy a gas has at a given moment. The heat that enters a gas might be zero ($Q=0$), if the container is thermally insulated, however, that does not mean that the temperature of the gas is zero.

Uses of Ultra filtration? 5 Marks

- Ultra filtration is mainly useful for fractionating peptides as well as the removal of proteins and other macromolecules based on their molecular size.
- UF is used extensively in the dairy industry; particularly in the processing of cheese whey to obtain whey protein concentrate (WPC) and lactose-rich permeate. In a single stage, a UF process is able to concentrate the whey 10-30 times the feed.
- Ultrafiltration (UF) is used to remove essentially all colloidal particles (0.01 to 1.0 microns) from water and some of the largest dissolved contaminants.

What is Anabolic Pathway? 3 Marks

Anabolic pathways consume energy to build complex molecules from simpler ones. The synthesis of protein from amino acids is an example of anabolism.

What is Forced dialysis? 3 Marks

Another method to concentrate samples is through forced dialysis. Vacuum is applied to a sample contained within a dialysis membrane; this effectively pulls" water, buffer salts and other low-MW compounds out of the dialysis sample chamber. Another form of dialysis involves pushing' samples through dialysis membrane by centrifugal force; this is the basis for protein concentrators, which have become popular in recent years.

What are the location of DNA In eukaryotes? 2 Marks

In other eukaryotes, types of such as plant and' as animals, mitochondria DNA and is found the chloroplasts in the nucleus, as well as in certain of plants.

Direct effect of ionizing radiation? 2 Marks

The direct effect of ionizing radiation and the radiant energy is immediately absorbed inside the cellular nucleus.

Define Blood Pressure? 2 Marks

Blood pressure (BP) is the pressure of circulating blood on the walls of blood vessels. Used without further specification, "blood pressure" usually refers to the pressure in large arteries of the systemic circulation.

What are Channelopathies? 2 Marks

Channelopathies are diseases that develop because of defects in ion channels caused by either genetic or acquired factors. Mutations in genes encoding ion channels, which impair channel function, are the most common cause of channelopathies.

State the equation for free energy of reaction? 3 Marks

All elements in their standard states (diatomic oxygen gas, graphite, etc.) have standard Gibbs free energy change of formation equal to zero, as there is no change involved. $\Delta_f G = \Delta_f G^\circ + RT \ln Q_f$, where Q_f is the reaction quotient.

What are the properties of Amino acids?

- They have a very high melting and boiling point.
- Amino acids are white crystalline solid substances.
- In taste, few Amino acids are sweet, tasteless, and bitter.
- Most of the amino acids are soluble in water and are insoluble in organic solvents.

What are the functions of amino acids?

A variety of roles in metabolism the building blocks of proteins forming parts of coenzymes as precursors for the biosynthesis of molecules such as heme.

What is HAZMAT?

HAZMAT is an abbreviation for “hazardous materials” substances in quantities or forms that may pose a reasonable risk to health, property, or the environment.

Nernst Heat Theorem?

Nernst stated that the entropy change for each chemical or physical transition between condensed phases, at temperatures very close to the absolute zero, is equal to zero. This statement is also referred to as the Nernst heat theorem. The statement of Nernst was simplified by Planck.